Shampoos

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A shampoo is a preparation of a surfactant (i.e. surface active material) in a suitable form – liquid, solid or powder – which when used under the specified conditions will remove surface grease, dirt, and skin debris from the hair shaft and scalp without adversely affecting the user

Formulation parameters

- viscous liquids
- clear or opaque
- viscosities 500–1500 centipoise
- pH 5.5
- Containing 20–40% solids

Formulation ingredients

Raw Materials

- Water
- Surfactants(Foam Boosters and Stabilizers)
- pH adjusters
- Viscosity modifiers
- Sequestering Agents
- Opacifiers
- Conditioning Agents
- Anti-dandruff Agents
- Perfumes
- Colors
- Preservatives

Water

This is the main ingredient in all shampoo preparations, comprising about 60-80% of the solution. It aids in diluting the cleaning agents, thereby reducing irritation. It makes the shampoo formula easier to spread on the hair and scalp. Surfactants are compounds that lower the interfacial tension of a between two phases. These are molecules that possess both hydrophillic and lipophillic moieties in their structure. They get adsorbed on the interface and helps the phases to misciblize.

1. Principal surfactants: Provide detergency and foam.

2. Secondary surfactants: Improve detergency, foam and hair condition.

Anionic surfactants are mostly used (good foaming properties). The hydrophilic portion carries a negative charge which results in superior foaming, cleaning and end result attributes.

• **Non-ionic surfactants** have good cleansing properties but do not have sufficient foaming power.
• **Cationic surfactants** are toxic and are hence not used. However, they may be used in low concentration in hair conditioners.
• **Ampholytics,** being expensive, are generally not used. However, they are mainly used as secondary surfactants and good hair conditioners.

SLS and (SLES)

Sodium Lauryl Sulfate (SLS) and Sodium Lauryl Ether Sulfate (SLES) play similar role in shampoos. SLS is a skin, eye and respiratory tract irritant. To make it less irritating, it is ethoxylated (by adding ethylene oxide), resulting in SLES. These agents are used to introduce gas bubbles into the water. The foam, also known as lather, is important, as it functions to spread the detergent over the hair and scalp, but it does not participate in cleaning.
• It is true that a shampoo applied to dirty hair will not foam as much as the same shampoo applied to clean hair. This is due to the sebum inhibiting bubble formation. Thus, a shampoo will foam less on the first shampooing and more on the second shampooing.
• Some of the prescription corticosteroid shampoos do not foam as much as cosmetic shampoos, but this does not mean their cleaning is inadequate.
• Examples: Lauroyl monoethanolamide, sarcosinates

pH Adjustors

The hair shaft consists of three layers, namely cuticle, cortex and medulla. Cuticle the outermost layer is made up of 7-10 layers of hard keratin. The Cortex, made up of hard keratin, constitutes approximately 80% of hair's total mass. The medulla is found in the center of the hair shaft. The hair follicle is a pocket in the skin from where the hair grows. The follicle grows through the epidermis and into the dermis. There are three main parts of the follicle: the Papilla, The Germinal Matrix, and the Hair Bulb. The papilla is found at the bottom of the follicle in the dermis. This is where the blood capillaries pass nutrients (food) and oxygen into the cells of the germinal matrix.New cells start the process of keratinization at hair bulb. Thus, hair shaft is a dead structure. it gets nutrition from the dermis. Hair cleansing preparations helps to make hairs look better but does not provide nutrition.

pH adjusters are used to prevent the hair shaft from alkalinization. Most detergents are having alkaline pH, which causes hair shaft swelling. This swelling loosens the protective cuticle predisposing the hair shaft to damage. Example: Citric acid, Glycolic acid.

Thickening Agents

These agents are used shampoo thick and creamy. Thickening may be achieved by adding salts or gums. Gums improve viscosity because of their gel-like properties.Eg.Tragacanth gum, Gum Karaya, Carboxy methyl cellulose.

How salts act as thickening agent for shampoos containing anionic surfactants.

The viscosity of the shampoo solution depends on the size and packing structure of micelles (tiny vesicles of surfactants).

• In general, higher charge density causes the micelles to repel and result in a thinner solution.

• The sodium ions from the salt lower the charge density of the micelle surface. This makes them more able to pack closer together and creates a thicker solution.

Opacifying Agents

Chemical agents added to the preparation to make it opaque, so that light does not pass through. These are usually added to give pearly shine, which offers no improve cleansing. It provides only optical effect e.g. Spermaceti, Alkanolamides of higher fatty acids, propylene , propylene glycol, Mg, Ca and Zn salts of stearic acid etc.

Conditioners

The conditioner functions to impart manageability, gloss, and antistatic properties to the hair. These are usually fatty alcohols, fatty esters, vegetable oils, mineral oils, orhumectants. Commonly used conditioning substances include hydrolyzed animal protein, glycerin, dimethicone, simethicone, polyvinylpyrrolidone, propylene glycol etc.

• Protein-derived substances are popular conditioners for damaged hair, as they can temporarily mend split ends. Split ends arise when the protective cuticle has been lost from the distal hair shaft and the exposed cortex splits. The protein derived substances holds the cortex fragments together until the next shampooing occurs.

Anti dandruff Agents

Medicated shampoos contain small amount of these actives, which are in contact with the scalp for only a short time. In order to be effective the active ingredient must work in the oil-water environment of the scalp and must be readily substantive to the scalp for continuing activity. Example: Selenium sulfide, zinc pyrithone, salicylic acid.

Perfumes

Shampoos include perfumes that are mostly concentrated. Example: Fruit fragrance

Colors

Used to impart color, different colors are used.

Preservatives

Shampoo formula containing water has the potential to be contaminated by pathogens. For this reason it is essential to include preservatives among shampoo ingredients, to prevent the growth of molds. Preservatives usually comprise only 0.1 –0.5%.

Manufacturing Process

Some agents are waxy solids at ambient temperature and require melting in a drum oven or similar before use. Demineralized water is most commonly used to minimize contamination of the product. No further processing is required after blending, and the product may be packed off directly from the mixing vessel.